

1. Protocol Synopsis & Overview

Protocol Title: A Phase I, Randomized, Double-Blind, Placebo-Controlled, Escalating Single-Dose Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of Compound Z in Healthy Adult Subjects.

Study Drug: Compound Z (10 mg, 30 mg, and 100 mg) **Phase:** I **Indication:** Healthy Volunteers (Translational Medicine / Neurology target)

2. Methods

2.1 Study Design and Dosing

This was a single-center, randomized, double-blind, placebo-controlled, single-ascending-dose (SAD) study. A total of 24 healthy subjects were enrolled into three sequential cohorts of 8 subjects each (randomized 6:2 to receive a single oral dose of Compound Z or matching placebo):

- **Cohort 1:** 10 mg (n=6 active, n=2 placebo)
- **Cohort 2:** 30 mg (n=6 active, n=2 placebo)
- **Cohort 3:** 100 mg (n=6 active, n=2 placebo)

2.2 Pharmacokinetic (PK) Sampling and Analysis

Blood samples for PK analysis were collected pre-dose (0 hours) and at 0.5, 1, 1.5, 2, 4, 6, 8, 12, 24, 36, and 48 hours post-dose.

Plasma concentrations of Compound Z were quantified using a validated liquid chromatography-tandem mass spectrometry (LC-MS/MS) method. PK parameters were calculated via non-compartmental analysis (NCA) using WinNonlin software, including:

- Maximum observed plasma concentration (C_{max})
- Time to reach C_{max} (T_{max})
- Area under the plasma concentration-time curve from time zero to infinity (AUC_{inf})
- Terminal elimination half-life (t_{1/2})

3. Pharmacokinetic Results

Compound Z was rapidly absorbed following oral administration, with median T_{max} ranging from 1.0 to 1.5 hours across all dose groups.

Plasma exposure (C_{max} and AUC_{inf}) increased in a cross-dose proportional manner from 10 mg to 100 mg. The mean terminal half-life (t_{1/2}) was determined to be approximately 12 to 14 hours, supporting a once-daily dosing regimen.

The mean PK parameters are summarized in the table below:

Parameter	10 mg (N=6)	30 mg (N=6)	100 mg (N=6)
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Cmax (ng/mL) , Mean (SD)	45.2 (6.8)	142.1 (21.4)	468.5 (55.1)
Tmax (h) , Median [Range]	1.0 [0.5–2.0]	1.25 [1.0–2.0]	1.5 [1.0–2.5]
AUCinf (ng·h/mL) , Mean (SD)	512 (72)	1,640 (215)	5,310 (680)
t1/2 (h) , Mean (SD)	12.1 (1.4)	12.8 (1.9)	13.5 (2.1)

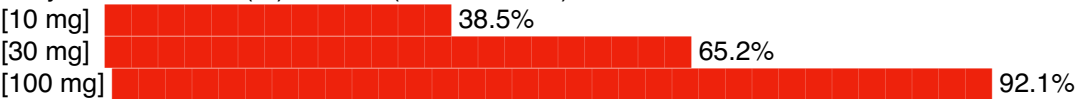
4. Pharmacodynamic (PD) Results

The target biomarker (Enzyme X activity) was measured in peripheral blood mononuclear cells (PBMCs) to evaluate target engagement.

Compound Z demonstrated a dose-dependent inhibition of Enzyme X activity.

- 10 mg Dose:** Induced a maximum mean inhibition of 38.5% at 2 hours post-dose, returning to baseline by 12 hours.
- 30 mg Dose:** Induced a maximum mean inhibition of 65.2% at 2 hours post-dose, with sustained inhibition (>30%) up to 24 hours.
- 100 mg Dose:** Achieved near-complete target inhibition, with a maximum mean inhibition of 92.1% at 2 hours post-dose. Sustained inhibition (>50%) was maintained through the 24-hour time point (p<0.01 compared to placebo).

Enzyme X Inhibition (%) at Peak (2h Post-Dose):



5. Summary and Next Steps

The pharmacokinetic and pharmacodynamic profiles of Compound Z following single ascending doses indicate linear, dose-proportional PK characteristics and predictable, dose-dependent target engagement.

The 100 mg dose achieved the target threshold of >80% biomarker inhibition deemed necessary for therapeutic efficacy. Based on these favorable PK/PD characteristics and the absence of dose-limiting toxicities, progression to a Multiple Ascending Dose (MAD) study in patients is recommended.